

Phase 1 report

Summary :

The dataset is first preprocessed by combining the **aspect term** and its corresponding **sentence** into a single input (e.g., "*Aspect: food Sentence: The food was amazing but the service was terrible.*"). This provides the model with both the aspect being evaluated and the context in which it appears. Since machine learning models cannot process raw text directly, the combined text is converted into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency)**. TF-IDF assigns higher weights to words that are more informative and distinctive while reducing the importance of very common words, creating a numerical vector that represents each text sample.

These TF-IDF feature vectors are then used to train a **Logistic Regression** classifier. During training, the model learns the relationship between the numerical features and the sentiment labels (negative, neutral, or positive). For prediction, new text undergoes the same TF-IDF transformation, and the trained Logistic Regression model uses the resulting feature vector to predict the most likely sentiment class for the given aspect. The model's performance is then evaluated using metrics such as accuracy, precision, recall, F1-score, and a confusion matrix.

1. Narrow Project Question

Research Question

Can a simple TF-IDF feature representation combined with Logistic Regression accurately classify the sentiment (Positive, Neutral, or Negative) of a specific aspect mentioned in student course feedback?

Motivation

Students often provide feedback that discusses multiple aspects of a course, such as the instructor, assignments, grading, or course materials. Instead of assigning one overall sentiment to an entire review, **Aspect-Based Sentiment Analysis (ABSA)** determines the sentiment expressed toward each individual aspect.

This project focuses on building a simple, reproducible baseline for this task rather than developing a production-ready educational assistant. The baseline will serve as a reference for more advanced methods in Phase 2.

2. Dataset Description

Dataset

The project uses a publicly available **Aspect-Based Sentiment Analysis (ABSA)** dataset containing course feedback.

Each sample consists of:

Column	Description
example_id	Unique identifier
text	Student feedback sentence
aspect_term	Aspect mentioned in the sentence
term_location	Position of the aspect within the sentence
class	Sentiment label

Example:

text	aspect_term	class
The lectures were excellent but the assignments were difficult.	lectures	1
The lectures were excellent but the assignments were difficult.	assignments	-1

The same sentence can therefore appear multiple times with different aspect terms.

Dataset Size

Number of samples:

3500+

Number of sentiment classes:

- Negative
- Neutral
- Positive

Original labels:

- -1 → Negative
- 0 → Neutral
- 1 → Positive

The labels are converted into:

Original	Model Label
-1	0
0	1
1	2

to simplify multi-class classification.

Why this dataset?

The dataset directly matches the research question because:

- every sample contains both a sentence and a specific aspect,
 - every aspect has a sentiment label,
 - it represents real student feedback,
 - it is suitable for supervised machine learning.
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3. *Baseline Setup*

Objective

Develop a simple and fully reproducible baseline for Aspect-Based Sentiment Analysis.

Pipeline

The baseline follows these steps.

Step 1: Load Dataset

The dataset is loaded from a CSV file using Pandas.

Step 2: Data Preparation

The model receives both the aspect and the sentence.

Instead of using only the review text, the input is constructed as

```
Aspect: assignments Sentence: The lectures were excellent but the assignments  
were difficult.
```

This allows the classifier to distinguish between different aspects mentioned in the same sentence.

Step 3: Label Conversion

The original labels

```
-1  
0  
1
```

are converted into

```
0  
1  
2
```

for compatibility with scikit-learn's multi-class classifier.

Step 4: Train/Test Split

The dataset is divided into

- 80% Training
- 20% Testing

using

- `random_state = 42`
- stratified sampling

to ensure reproducibility and preserve the class distribution.

Step 5: TF-IDF Feature Extraction

Text is converted into numerical vectors using TF-IDF (Term Frequency–Inverse Document Frequency).

Configuration:

- `lowercase=True`
- English stop-word removal
- unigram + bigram features
- maximum vocabulary = 10,000
- minimum document frequency = 2

To avoid data leakage, the TF-IDF vectorizer is fitted only on the training data and then applied to the test set.

Step 6: Logistic Regression

A Logistic Regression classifier is trained using:

- `solver = liblinear`
- `max_iter = 3000`
- `class_weight = balanced`
- `random_state = 42`

Class balancing helps reduce bias toward majority classes.

Step 7: Prediction

The trained model predicts the sentiment class for each sample in the test set.

Step 8: Evaluation

The model is evaluated using multiple classification metrics and a confusion matrix.

Step 9: Demo

A small demonstration predicts the sentiment of unseen sentences for different aspects.

Example:

Sentence:

The food was amazing but the service was terrible.

Aspect:

food

Prediction:

Positive

Aspect:

service

Prediction:

Negative

This demonstrates that the model considers both the sentence and the specified aspect.

4. Evaluation Plan

The model is evaluated using Accuracy, Precision, Recall, Macro F1-score, a detailed classification report, and a confusion matrix.

- **Accuracy** – measures the overall proportion of correctly classified samples.
 - **Precision** – measures the proportion of predicted samples for each class that are correctly classified.
 - **Recall** – measures the proportion of actual samples of each class that are correctly identified.
 - **Macro F1-score** – computes the F1-score for each sentiment class separately and then averages them equally, making it suitable for evaluating performance on potentially imbalanced datasets.
 - **Classification Report** – provides per-class Precision, Recall, F1-score, and the number of samples (support) for each sentiment class.
 - **Confusion Matrix** – summarizes the model's predictions by showing the number of correctly and incorrectly classified samples for each class, making it easier to identify common classification errors.
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5. Planned Error Analysis (Phase 2)

During Phase 2, the following error categories will be investigated.

1. Confusion Between Neutral and Positive

Neutral feedback may contain mildly positive or mildly negative language, making it difficult for the classifier to distinguish between neutral and polar sentiments.

2. Multiple Aspects in One Sentence

Some sentences contain different sentiments toward different aspects.

Example:

The lectures were excellent but the assignments were difficult.

The classifier may incorrectly predict the sentiment if it fails to focus on the specified aspect.

3. Short or Ambiguous Feedback

Very short comments such as

"Average."

or

"Fine."

provide little contextual information, making sentiment prediction more difficult.

6. Reproducibility

The baseline is fully reproducible.

Reproducibility is ensured through:

- fixed random seed (`random_state = 42`)
- stratified train/test split
- fitting TF-IDF only on training data
- publicly available Python libraries:
 - Pandas
 - scikit-learn

Any user with the same dataset can reproduce the reported results by running the provided script.

7. Expected Outcome

This baseline establishes a simple and interpretable reference model for Aspect-Based Sentiment Analysis on course feedback.

Although more advanced transformer-based models (e.g., BERT or RoBERTa) are expected to achieve higher accuracy, the TF-IDF + Logistic Regression approach offers several advantages:

- simple implementation,
- fast training,
- low computational requirements,
- easy reproducibility,
- strong baseline for comparison in future work.

The results obtained from this baseline will serve as a benchmark for evaluating more sophisticated models in Phase 2.